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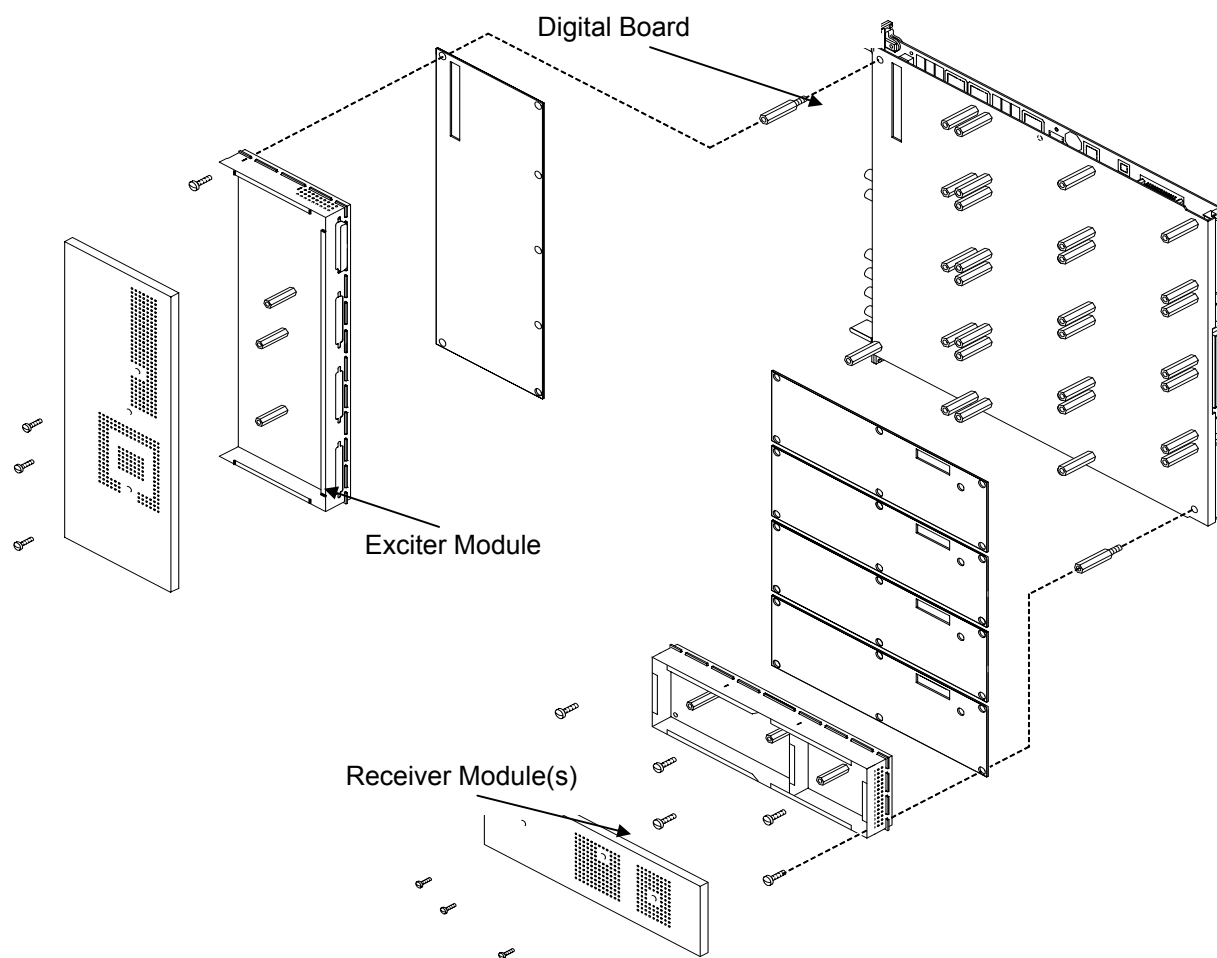
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1- CERD ASSEMBLY BOARD

The CERD (Combined Exciter/Receiver Digital) Board is located in the ISE/TPS Chassis. Refer to Illustration 1-1 for the CERD Assembly.

Note

The assembly shows standoffs on the Digital Board. Please refer to Service Note 60955 for a description.

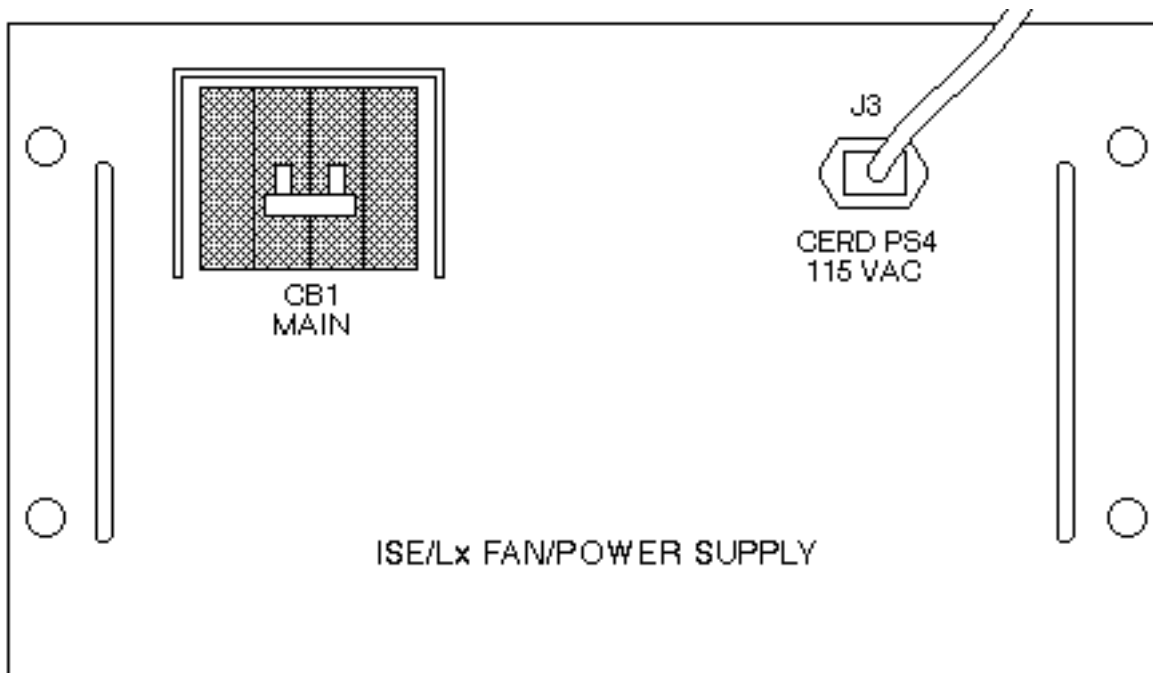


M4512A

CERD ASSEMBLY
ILLUSTRATION 1-1

1-1 Removal of CERD Assembly

1. Remove the System Cabinet front cover.
2. Turn off the main circuit breaker, CB1 (see Illustration L4601A).



ISE/TPS FAN/POWER ASSEMBLY - FRONT VIEW
ILLUSTRATION L4601A

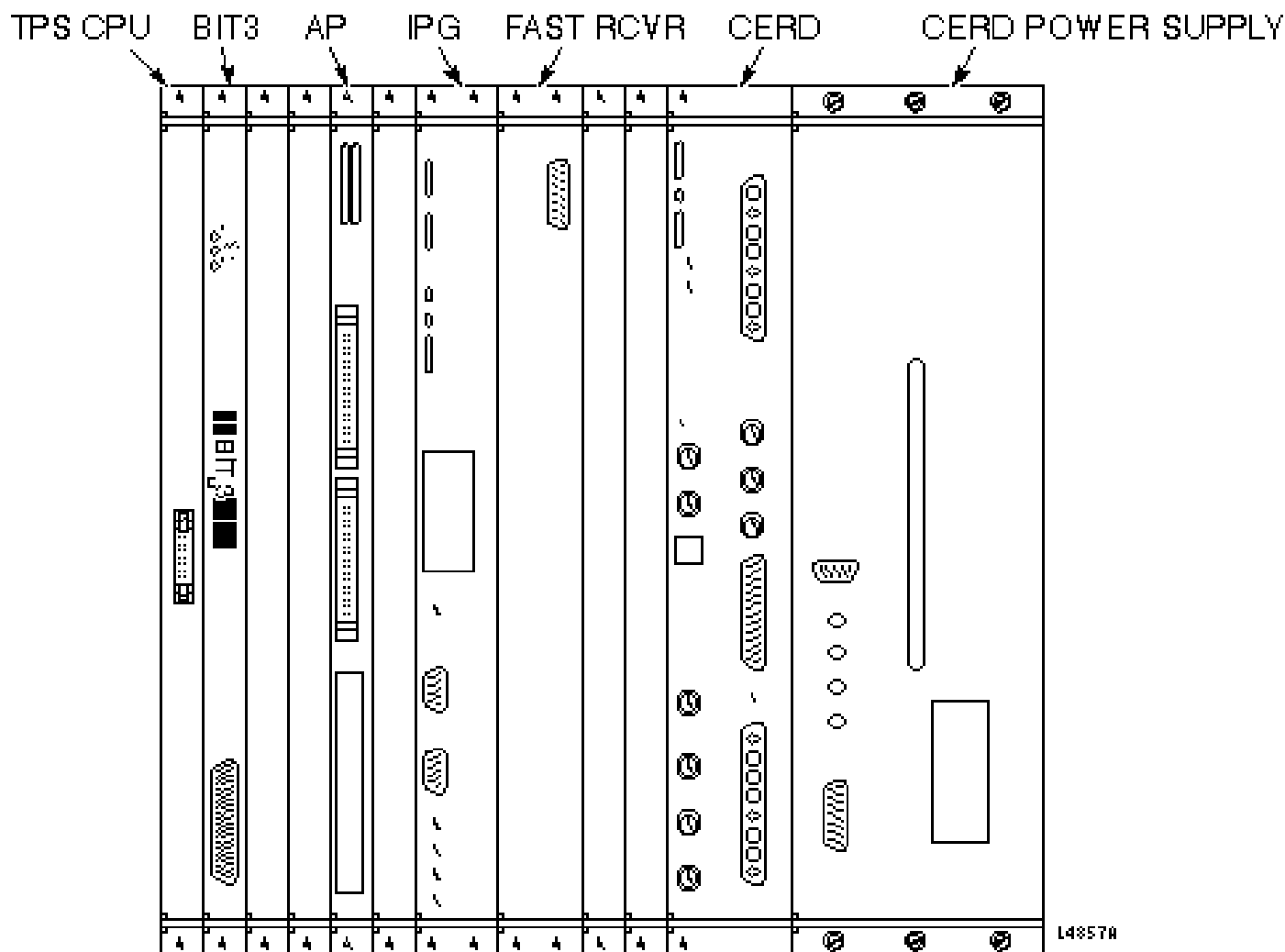
CAUTION

Equipment damage possibility. Loosen the two front screws from the blank panel to the left (slot # 12) of the CERD Assembly and Remove the Blank panel. Components and RF fingers (EMC) on the CERD can be damaged if the blank panel is not removed.

CAUTION

Equipment damage possibility. ESD precautions must be used when removing the CERD assembly or replacing any modules on the CERD.

3. Loosen the two front panel screws. Remove the CERD Assembly from the ISE/TPS Chassis (see Illustration L4729A).



ISE/TPS CHASSIS - FRONT VIEW
ILLUSTRATION L4729A

4. Remove the appropriate modules from the CERD Assembly.

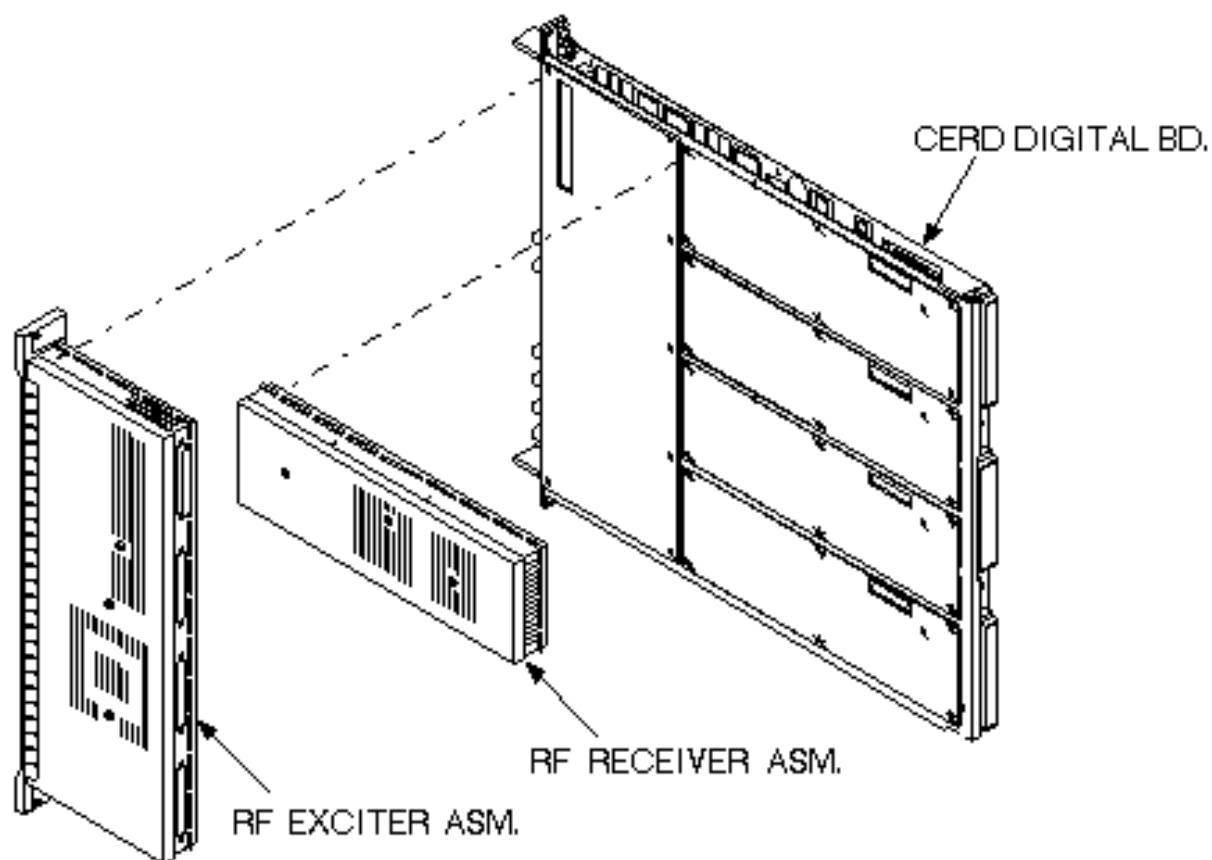
1-1-1 RF Receiver Module(s)

Note

When removing Receiver module(s), the existence of the Digital Board standoffs now requires the following:

To remove the Receiver module(s), the cable to the receiver has to be disconnected and pulled back into the Exciter. so that the Receiver(s) clear the Digital Board standoffs.

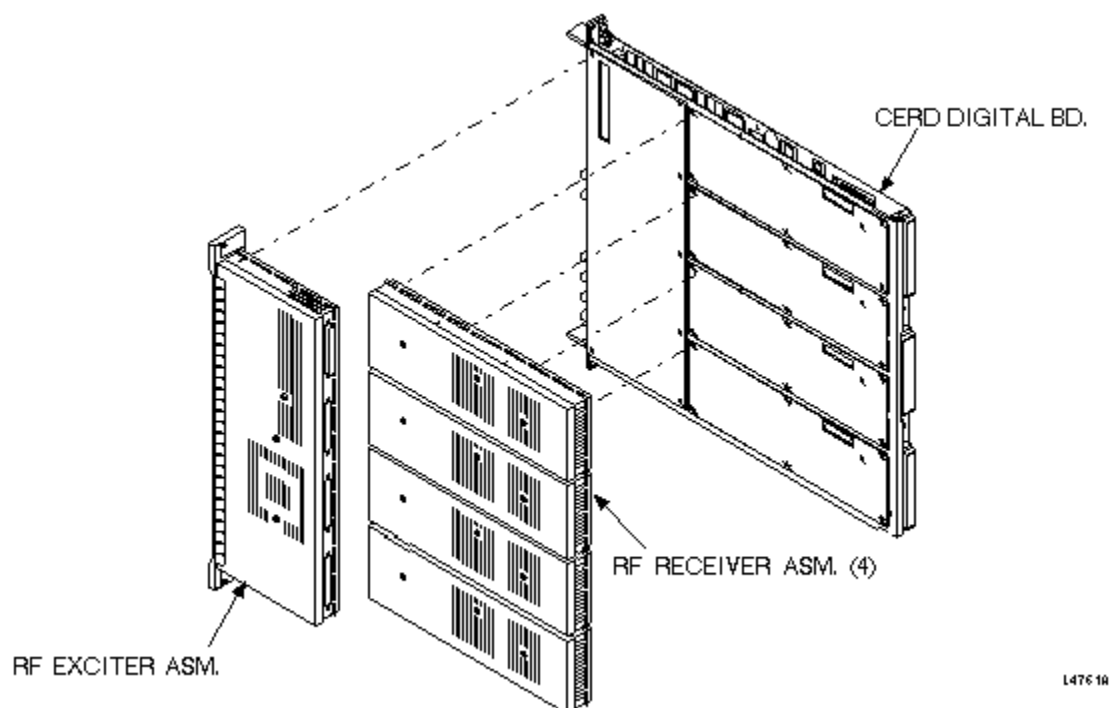
See Illustration L4760a for a single-channel CERD Assembly which includes only 1 receiver module.



L4760A

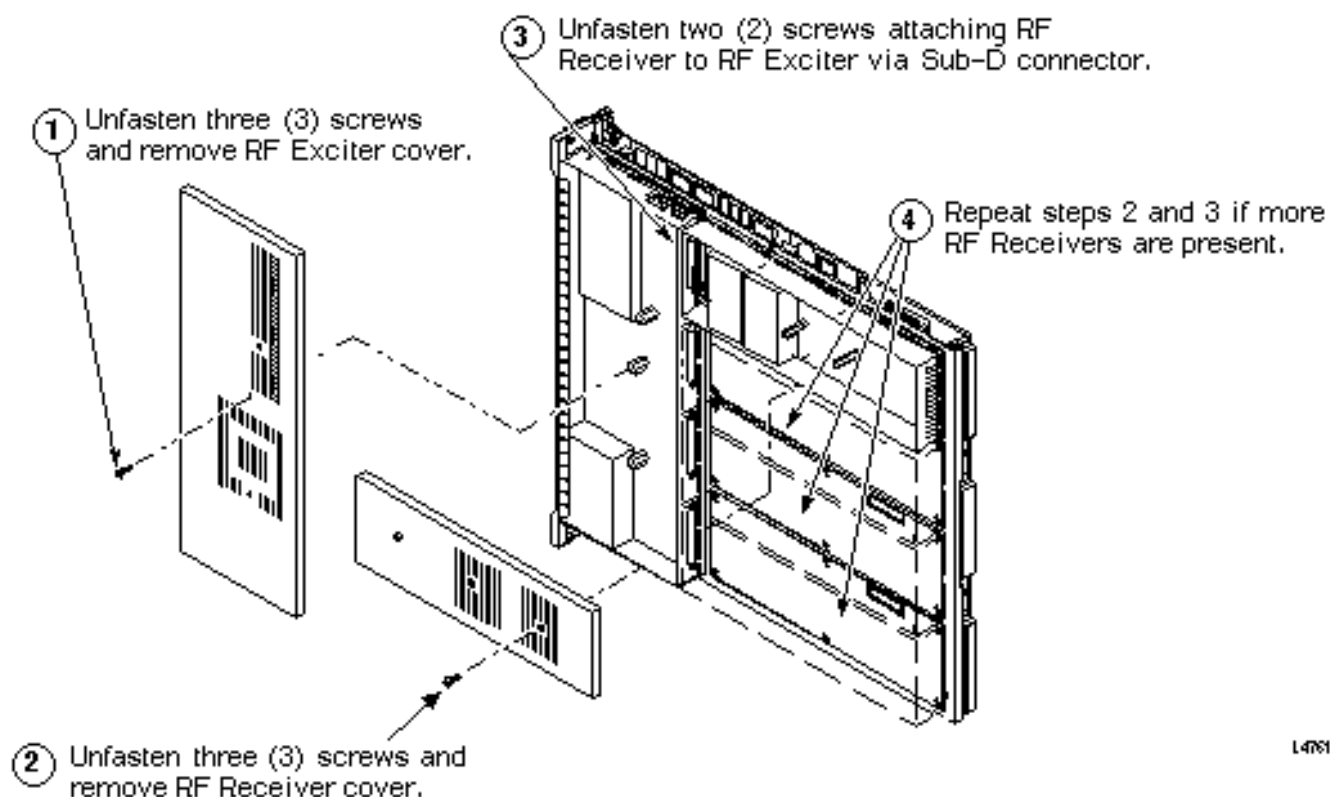
SINGLE-CHANNEL CERD REPLACEMENT
ILLUSTRATION L4760A

See Illustration L4761A for a four-channel CERD Assembly which includes four receiver modules.



FOUR-CHANNEL CERD REPLACEMENT
ILLUSTRATION L4761A

If you are replacing or removing one or more of the RF Receiver Modules See Illustration L4761B.



REMOVAL OF RF RECEIVER(S) FROM CERD
ILLUSTRATION L4761B

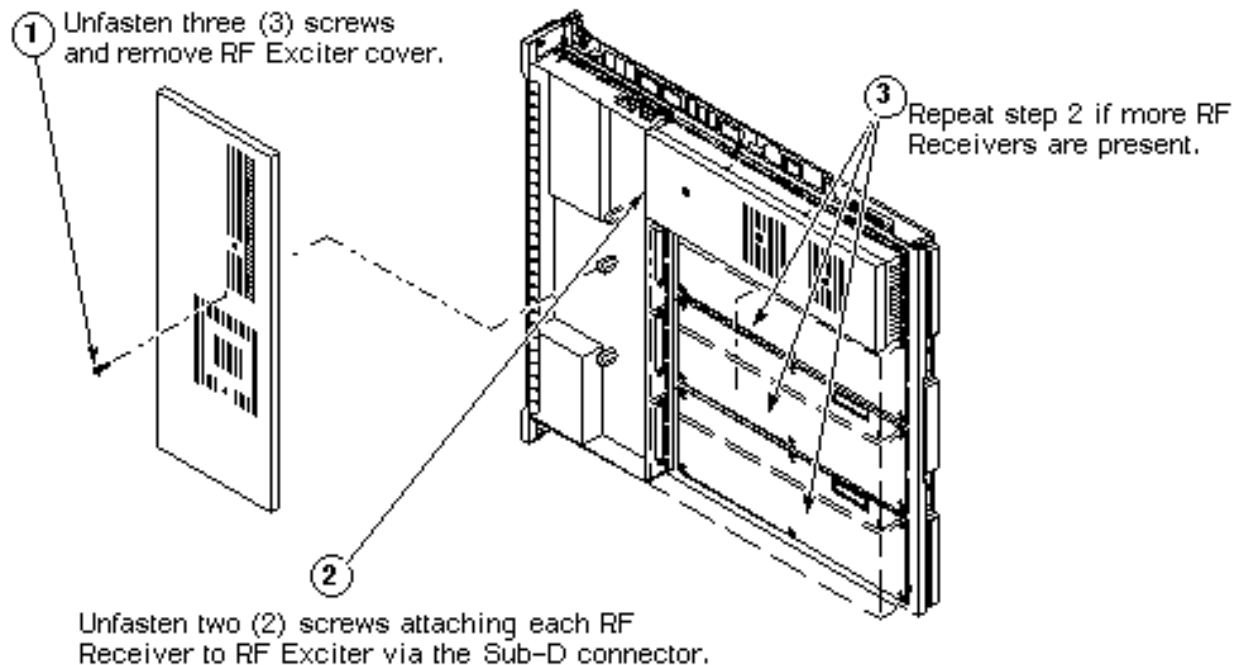
1-1-2 RF Exciter Module

Note

When removing the Exciter module, the existence of the Digital Board standoffs now requires the following:

To remove the Exciter, the Receiver(s) must be removed first so that the Exciter can be lifted up to clear the Digital Board standoffs.

If you are replacing or removing the RF Exciter module see Illustration L4761C.

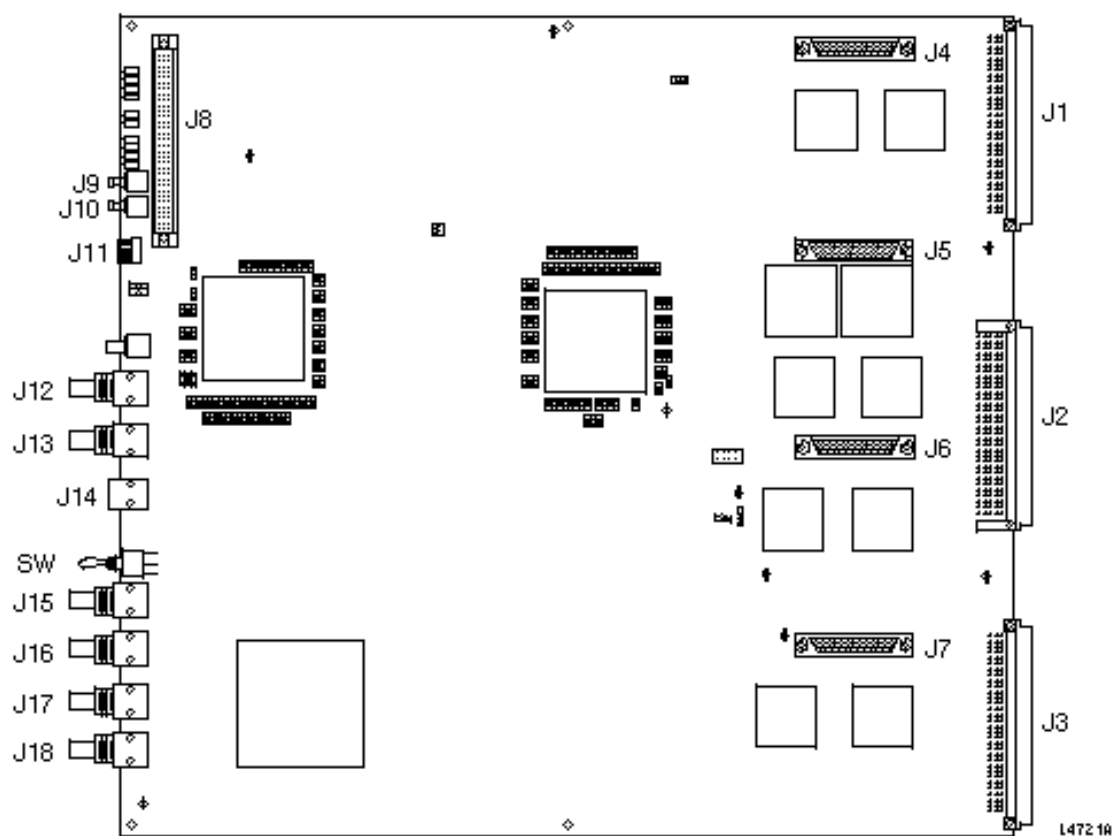


L4761B

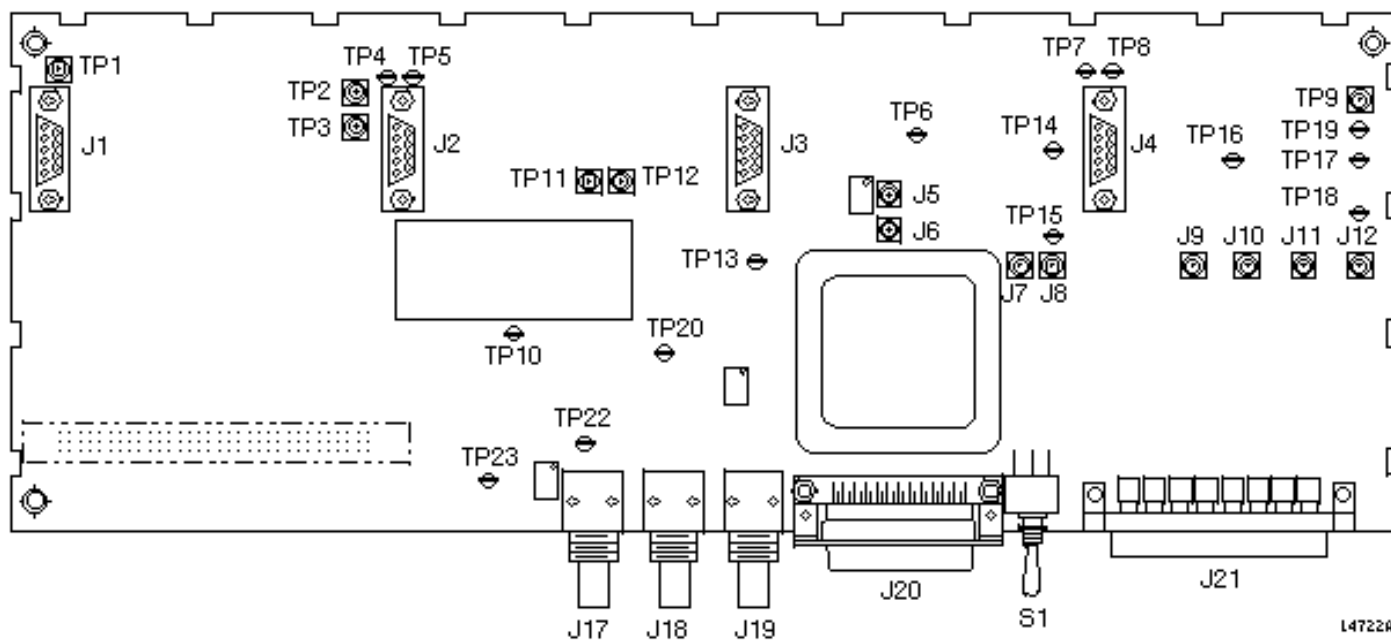
REMOVAL OF RF EXCITER FROM CERD
ILLUSTRATION L4761C

1-2 CERD Assembly Setup

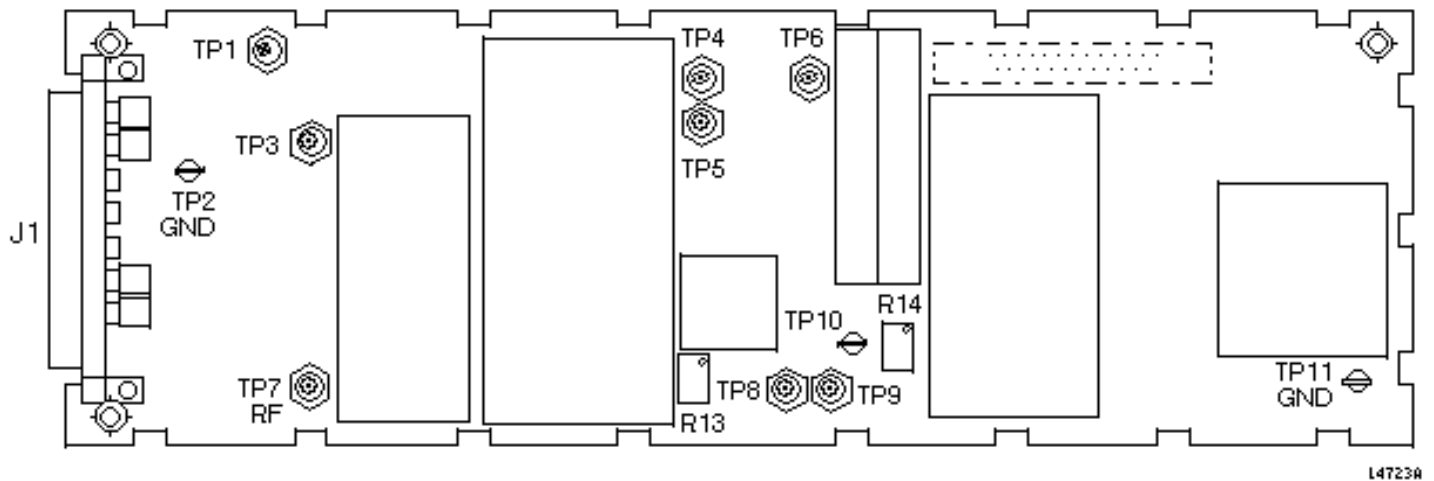
Illustration L4721A shows the CERD Assembly, Illustration L4722A shows the CERD Exciter Board (2148300 and 2148300-2), and Illustration L4723A shows the Modular Receiver Card.



CERD ASSEMBLY
ILLUSTRATION L4721A



CERD EXCITER BOARD (2148300 AND 2148300-2)
ILLUSTRATION L4722A



MODULAR RECEIVER CARD
ILLUSTRATION L4723A

Refer to Table 1 for Switch information.

TABLE 1
SWITCH INFORMATION

Switch	Type	Comments
<i>Digital Board Switches:</i>		
CERD Reset	Push-button	Treated by CERD same as VME backplane reset, local to CERD only
Rho/Theta/Omega Sample Select	DPDT, center off	Three positions corresponding to RHO, THETA, or OMEGA selected to be output on the "RHO/THETA/OMEGA Sample" connector
<i>RF Exciter Board Switches:</i>		
RF Exciter Output Toggle Disable		Disconnect condition = RF output disabled

Refer to Table 2 for jumper information.

TABLE 2
JUMPER INFORMATION

Jumper	Board Assembly 46-317038	Jumper position	Notes
JP1		Set to position A (pins 1&2) to enable TBC and disable emulator connector.	
		Set to position B (pins 2&3) to enable emulator connection and disable TBC.	
JP3		Normally set to A position. Set to B position to enable FLASH boot block to be reprogrammed.	

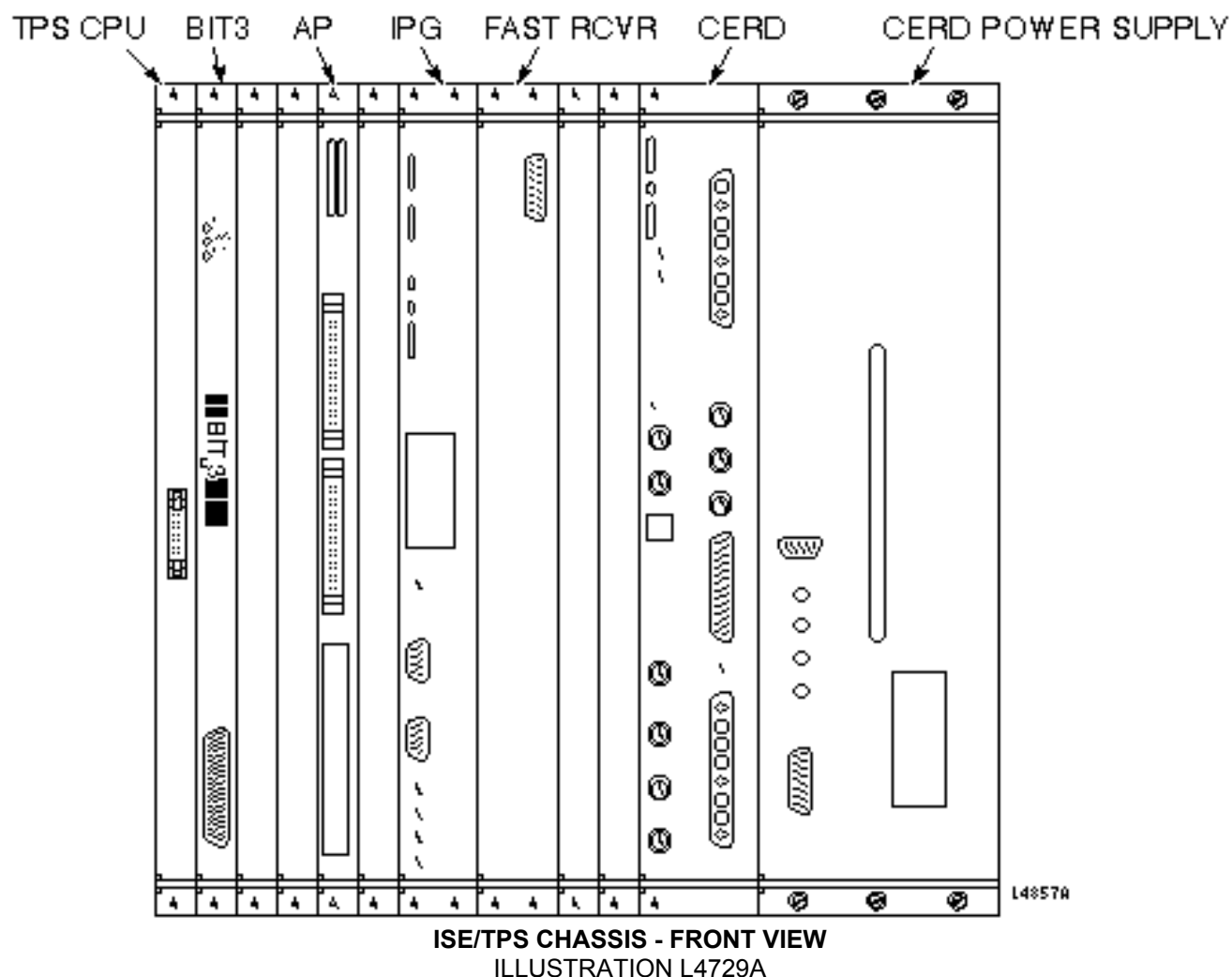
The functions of the LEDs on the Digital Board only are in Table 3 - LED information.

TABLE 3
LED INFORMATION

LED	Color	Description	Notes
LO1	Green	On when LO1 is present on all installed receivers	
LO2	Green	On when Exciter LO2 is present	
RCVR UNBLK	Green	On when Receiver #1 is unblanked	
EX UNBLK	Green	On when Exciter is unblanked	
CAP	Green	Connected to input line to CAP watchdog. Toggled under software control a minimum of once per second (0.5 Hz flash rate minimum)	
ASC	Green	Connected to input line to ASC watchdog. Toggled under software control a minimum of once per second (0.5 Hz flash rate minimum)	
VME	Green	On when slave select signal asserted out of VIC068A. Pulse stretched as required for visibility	
VSB	Green	On during VSB activity. Pulse stretched as required for visibility	
SSP	Green	On during SSP writes to the CERD. Pulse stretched as required for visibility	
XD	Green	On during XD activity. Pulse stretched as required for visibility	

1-3 Installation of CERD Assembly Board

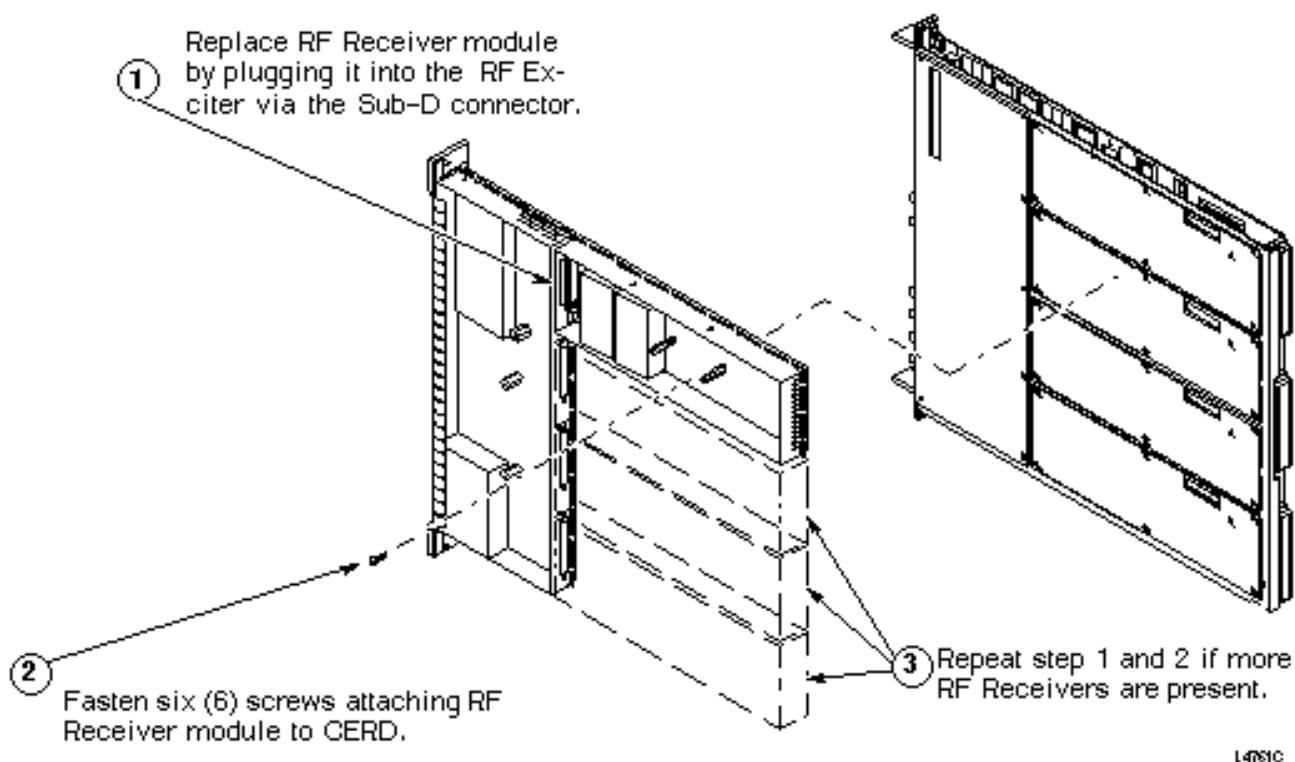
1. Install the appropriate module(s) into the CERD Assembly (refer to sections 1-3-1 and 1-3-2)
2. Install the CERD Assembly in the ISE/TPS Chassis and tighten the two front panel screws (see Illustration L4729A).



3. Turn on the main circuit breaker, CB 1 (see Illustration L4601A in Section 1-1).
4. Install the System Cabinet front cover.
5. If Signa Horizon (8.X) software is running, **[Reset TPS]**.

1-3-1 RF Receiver Module(s)

See Illustration L4760A in Section 1-1 for a single-channel CERD Assembly.
See Illustration L4761A in Section 1-1 for a four-channel CERD Assembly.
If you are replacing one or more of the RF Receiver modules see Illustration L4761D.

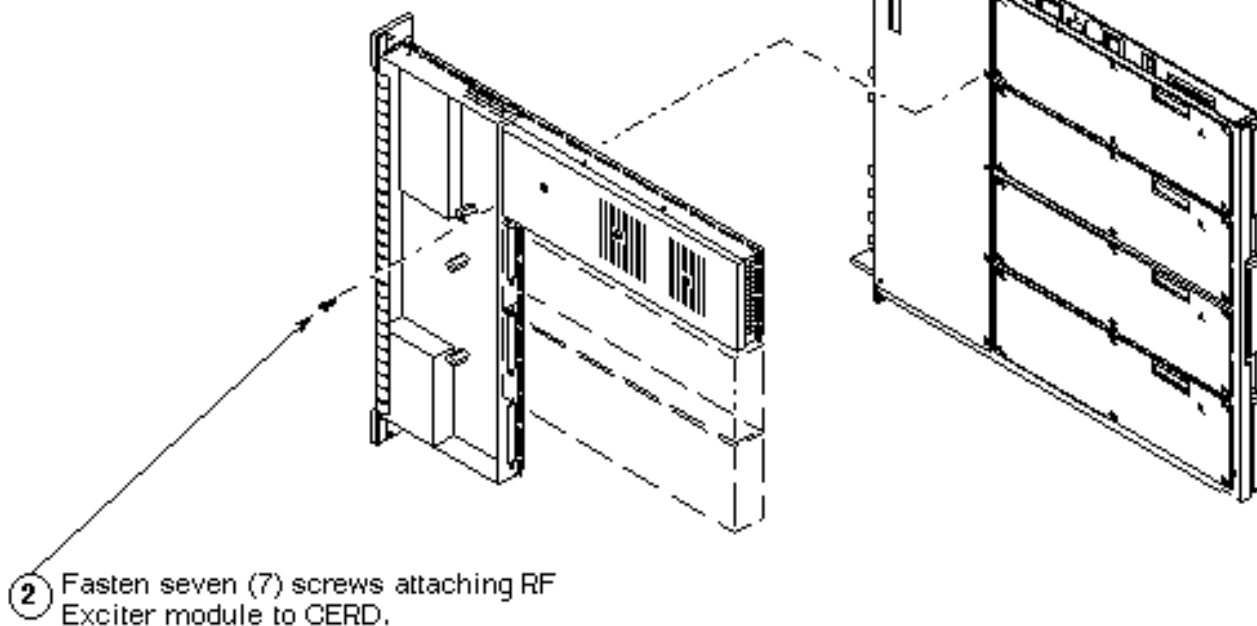


INSTALLATION OF RF RECEIVER(S) ONTO CERD
ILLUSTRATION L4761D

1-3-2 RF Exciter Module

If you are replacing the RF Exciter module see Illustration L4761E.

- ① Replace RF Exciter module by plugging it into the Receiver(s) on the CERD.



L4761C

INSTALLATION OF RF EXCITER ONTO CERD
ILLUSTRATION L4761E

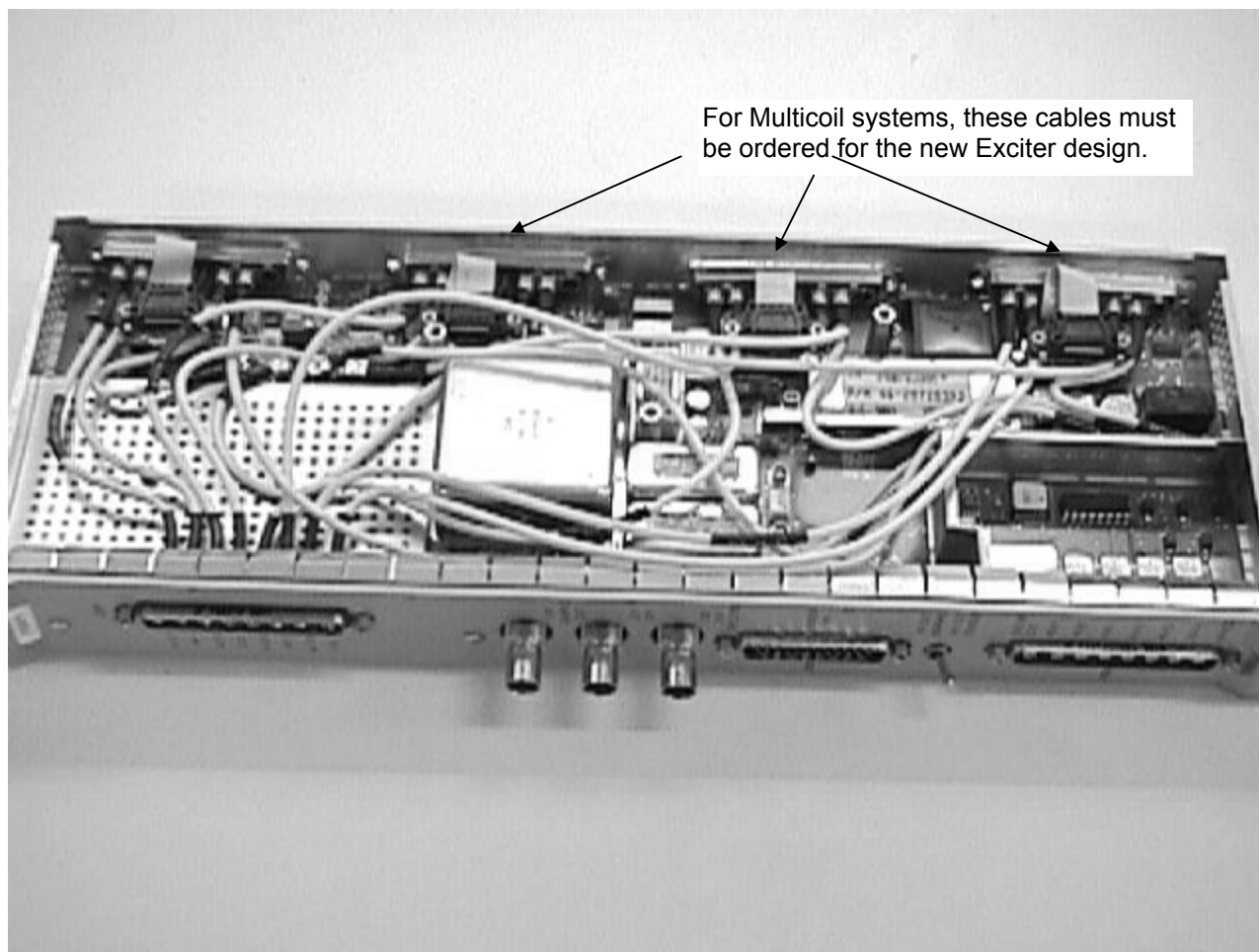
Note

If you receive a board with the part number of 2148300-5 or 2148300-6 and the site has multicoil, then a new cable kit is also required (2210553).

Note

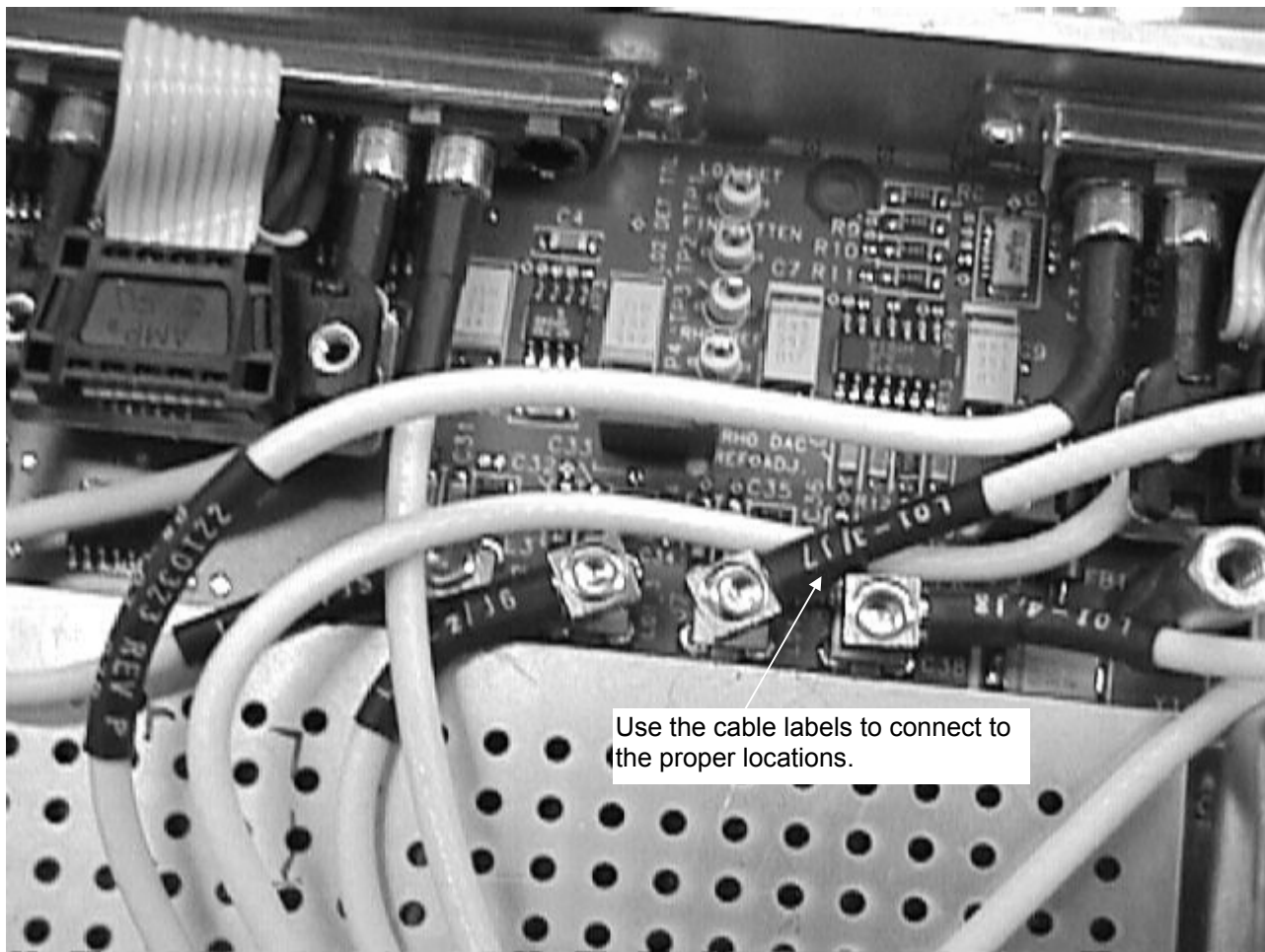
If replacing the Exciter board with 2148300-5 or 2148300-6, the probe filter must be removed. The filter has been incorporated into the new module.

Refer to Illustration 1-2 for board layout and the cabling locations for the 2148300-5 and 2148300-6 Exciters.



EXCITER BOARD
ILLUSTRATION 1-2

Illustration 1-3 is an example of the new connector locations on the Exciter board (2148300-5 and 2148300-6). To install the multicoil cables, insert the DB25 connector into the connector cutout and connect the cables to the proper connections using the labels provided on the cables.



CONNECTOR LOCATION DETAILS - EXCITER BOARD 2148300-5 AND 2148300-6
ILLUSTRATION 1-3

1-4 Functional Checks Required

1. Allow thirty minutes for the RF Exciter to warm up.
2. Connect the Exciter output, (MR2A7J1) to the oscilloscope 50-ohm input. For high Z oscilloscope input, use 50 ohms through termination. See note below for connection point.
3. At the operator work space, prepare the system for a **APB Cal, series 1: Body** scan using the *Service Protocols* procedure, or for the alternate proprietary procedure, see the following:

This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

- a. (If you haven't already done so:)

Click on **[New Pt]**, and enter

Id: **geservice**

Name: **apb cal body**

Weight (Lb): **111**

Set Patient Protocols to **Service**.

- b. Depending on your Release,

8.1: Select **[other]** to go to the protocol list on the system. Then select the protocol by name: **APB Cal**. Select series 1: **Body**.

8.2: In the protocol field, type **o.41.1** (o=Other, 41 =series) to load the protocol.

- c. **[Save Series]**.

- d. Select **[Research Options]** with the mouse cursor, and right click. Then select **[Setup Params]**. Set TG to **50** (transmit gain=90dB).

In the next steps, you will change the CV values as indicated. Depending on which Release you are using, see the User Interface Tutorial for specific help with User CVs.

- e. Set value of CV calmode to **5** (Dual Logamp Waveform).

- f. Set value of CV ia_rf1 to **32766** (sets 90° pulse full scale).

- g. Set value of CV ia_rf2 to **0** (turns off 180° pulse).

- h. Select **[Download]** under **[Research Options]**, and **[Manual Prescan]**.

If you need help, see the User Interface Tutorial.

4. **[Manual Prescan]**. Set TG = **200**. Measure the Exciter output to make sure its peak RF power is within specification on an error free oscilloscope with input termination of 50 ohms), see Table 6.

TABLE 6
SPECIFICATIONS WITH AND WITHOUT PROBE FILTER

Condition	Specification	
	Min	Max
RF OUT measured at the CERD Board:	893mVPP(+3dBm)	1.1 VPP (almost +5dBm)
RF OUT measured at MR2A11J1 without PROBE Filter:	796mVPP(+2dBm)	1.0 VPP (almost +4dBm)
RF OUT measured at MR1A20A1J3 without PROBE Filter and cable losses:	632mVPP(0dBm)	893mVPP (almost +3dBm)
RF OUT measured at MR2A11J1 with external PROBE Filter:	502mVPP(-2dBm)	709mVPP (+1dBm)
RF OUT measured at MR1A20A1J3 with external PROBE Filter and cable losses:	399mVPP(-4dBm)	632mVPP (0dBm)

5. Reconnect RF output cable to inside of J-1 cable interface.

Note

This measurement should be taken at the end of the exciter output cable. The connection cable may have 0.4dB/10 feet loss. If the level tested below 0dBm, this usually means something is wrong with the exciter. Some old versions of the exciter may normally output lower than 0dBm. The APM board can handle exciter output as low as -4dBm ($V_p=0.20V$). if you are confident about the exciter, you can still use the lower output (-4 to 0dBm) exciters.

Note

The connection cable used between the exciter and RF in (J105) of the APM is a sensitive part. Be sure to use double shielded cable to reduce noise collection that would possibly affect MR image quality.

6. Perform procedure for EPI Bandpass Asymmetry Correction Characterization.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	July 21, 1998	K. L-Parker	Added new Exciter board details
2	May 20, 1999	S.M.Atladottir	Updated Procedure References for New GUI
3	June 27, 2003	Hawthorne	Added step 6 in section 1-4 functional checks required to do a BACC after installation of CERD Board.